A) Working Principle

In a lead acid cell, sulphuric acid is used as an electrolyte. In this H_2So_4, electrolyte is poured after pouring water in it. Then, sulphuric acid dissolves and the molecules of hydrogen and sulphate are formed. In this, hydrogen ions are positive and sulphate ions are negative.

Two electrodes of battery are dipped in an electrolyte and DC supply is given as an input. Hydrogen positive ions go towards negative plate of electrode. Sulphate negative ions go towards positive plate of the electrode. In this way lead acid battery functions.

The sign(+) indicates positive terminal and sign(–) indicates negative terminal

Positive terminal-17.5mm dia

Negative terminal-16mm dia

b) Chemical Reactions During Discharging

When the cell is discharging, current flow in the external circuit is from positive to negative. (See fig. 4.8) The flow of current through the electrolyte (H_2SO_4) splits into positive hydrogen ion (H_2^+) and two negative sulphate ions (SO_4^{-2}).

Each sulphate ions move towards the cathode and on reaching there, give up two electrons to become radical SO_4, attack the metallic lead cathode and form lead sulphate, whitish in colour according to the chemical equation.

At Anode, H_2 combines with oxygen of PbO_2 and H_2SO_4 attacks lead to form $PbSO_4$.

At Anode:

$$PbO_2 + H_2 + H_2SO_4 \rightarrow PbSO_4 + 2H_2O$$

At cathode: $Pb + SO_4 \rightarrow PbSO_4$

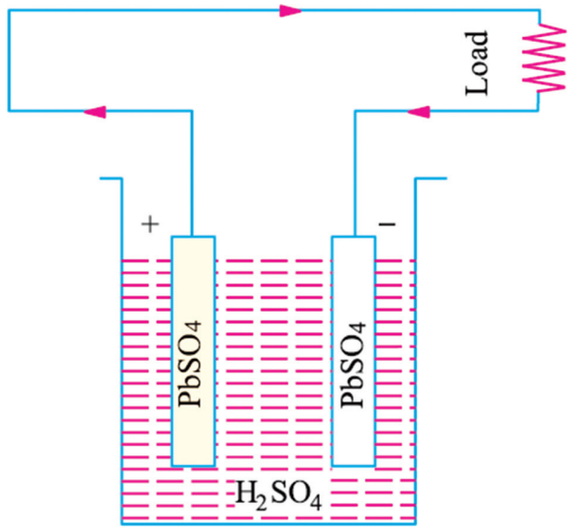


Fig 4.8 *Discharging Process*

Physical Changes While Discharging

1. Both the positive and negative plates are slowly converted into lead sulphate $PbSO_4$ (white in colour)
2. Water is formed during discharge. So the acid becomes more and more dilute. Specific gravity of sulphuric acid solution decreases.
3. Decrease in emf

c) Chemical Reaction During Charging

For recharging, the anode and cathode are connected to the positive and the negative terminal of the DC main supply.The hydrogen ions are positively charged move towards the cathode. (as in fig 4.9)

Sulphate ions move to the anode, and the following chemical reaction occurs.

$$\text{At Anode: } PbSO_4 + H_2 \rightarrow Pb + H_2SO_4$$

At Cathode:

$$PbSO_4 + SO_4 + 2H_2O \rightarrow PbO_2 + 2H_2SO_4$$

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